Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

Journal of Occupational and Organizational Psychology (2010), 83, 955–979
© 2010 The British Psychological Society

The British Psychological Society

www.bpsjournals.co.uk

The impact of methodological moderators on prevalence rates of workplace bullying. A meta-analysis

Morten Birkeland Nielsen*, Stig Berge Matthiesen and Ståle Einarsen

Department of Psychosocial Science, University of Bergen, Norway

The aim of this study was to investigate how different measurement methods and sampling techniques contribute to the observed variation in prevalence rates of workplace bullying. A total of 102 prevalence estimates of bullying from 86 independent samples ($N = 130,973$) were accumulated and compared by means of meta-analysis. At an average, the statistically independents samples provided an estimate of 14.6%. Yet, the findings show that methodological moderators influence the estimated rates. As for measurement method, a rate of 11.3% was found for studies investigating self-labelled victimization from bullying based on a given definition of the concept, whereas a rate of 14.8% was found for behavioural measure studies, and 18.1% for self-labelling studies without a given definition. A difference of 8.7% points was found between randomly sampled and non-randomly sampled studies. When controlling for geographical differences, the findings show that geographical factors also influence findings on bullying. Hence, findings from different studies on workplace bullying cannot be compared without taking moderator variables into account.

During more than 20 years of research on workplace bullying, a wealth of studies on the occurrence of the phenomenon has been conducted (see Einarsen, Hoel, Zapf, & Cooper, 2003a). In these studies, bullying has been found to be a widespread problem in contemporary working life (see Nielsen *et al.*, 2009; Zapf, Einarsen, Hoel, & Vartia, 2003, for an overview). In Great Britain, for example, every 10th employee reports exposure to bullying at work (Hoel, Cooper, & Faragher, 2001). Yet, estimates have been found to vary extensively both between and within countries (Nielsen *et al.*, 2009; Zapf *et al.*, 2003). The prevalence rates in Denmark and Turkey may be taken as an example. While a prevalence of 51% has been established in Turkey (Bilgel, Aytac, & Bayram, 2006), a rate of less than 3% has been found in Denmark (Agervold, 2007). Yet, prevalence rates also vary within countries, as another Danish study reported a rate of almost 13% (Høgh & Dofradottir, 2001).

*Correspondence should be addressed to Morten Birkeland Nielsen, Department of Psychosocial Science, University of Bergen, Christies Gate 12 (4 etg), 5055 Bergen, Norway (e-mail: morten.nielsen@psysp.uib.no).

DOI:10.1348/096317909X481256

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

Although several factors, such as cultural characteristics and societal changes, may contribute to explain variations in prevalence rates, issues related to research methodology must also be taken into consideration (Nielsen *et al.*, 2009). As studies on workplace bullying have utilized a variety of different measurement methods, instruments, and study designs (Agervold, 2007; Zapf *et al.*, 2003), it is reasonable to assume that methodological artifacts influence observed prevalence rates. Hence, it is difficult to determine from the literature with any degree of certainty the percentage of employees who have experienced workplace bullying. As we will show in this paper, attempts to derive such basic information are obstructed by important methodological problems shared by many studies. A resolution of these problems is critical in order for social scientists to have a meaningful impact on legal and policy-related issues. Furthermore, reliable and valid methods to assess workplace bullying are important for the development and implementation of effective intervention strategies to prevent workplace bullying (cf. Schat, Frone, & Kelloway, 2006). For instance, governments and organizations are dependent on correct estimates of bullying in order to know how much time and resources that is needed to handle the problem. Precise estimates are also important in research, as over or underestimation of the occurrence of bullying may lead to Types I and II errors.

By using a meta-analytic design, the aim of this study was to investigate how methodological issues contribute to the variance in the reported prevalence rates of workplace bullying. More specifically, the study's first objective was to provide some reasonable cumulative estimates, given the limitations of the research literature, of the percentage of employees who have experienced workplace bullying. The second objective was to examine the impact of measurement method and sampling procedure on the observed prevalence rates of bullying.

The concept of workplace bullying

Several studies have shown that workplace harassment and bullying are problems with significant effects on targets, observers, and organizations (cf. Einarsen & Mikkelsen, 2003; Rayner, Hoel, & Cooper, 2002). As a telling example, two recent meta-analyses that compared the impact of psychological aggression and sexual harassment on outcome variables, such as job-satisfaction, stress, anger, and health, found that the effects of psychological aggression and bullying were significantly more severe than the effects of sexual harassment (Hershcovis & Barling, 2009; Lapierre, Spector, & Leck, 2005).

As a concept, workplace bullying is a collective expression that includes various forms of mistreatment and hostile behaviour in the workplace setting (Fox & Stallworth, 2005). Although many different definitions of workplace bullying have been proposed (cf. Einarsen, Hoel, Zapf, & Cooper, 2003b; Keashly & Jagatic, 2003; Rayner *et al.*, 2002), the majority resembles Einarsen's (2000, pp. 383–384) clarification of the concept which again is building on Olweus (1991) definition of bullying among schoolchildren: '… bullying refers to situations where a person repeatedly and over a period of time is exposed to negative acts (i.e., constant abuse, offensive remarks or teasing, ridicule or social exclusion) on the part of co-workers, supervisors or subordinates, and where the person confronted have difficulties defending himself/herself against the mistreatment'.

According to this definition, workplace bullying is characterized by four main criteria, with the first being that the target is exposed to direct or indirect, but highly

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

unwanted, negative acts of a non-sexual and mainly non-violent nature. These acts may range from subtle, even unconscious, incivilities to blatant, intentional emotional abuse (Fox & Stallworth, 2005). The second criterion is that the negative acts in question are repeated regularly. Bullying, therefore, is not about isolated episodes or events, but rather about systematic aggressive behaviour that is repeatedly directed towards one or more employees. The third criterion is that the negative events take place over a prolonged time period (Einarsen, Matthiesen, & Hauge, 2008), whereas the last characteristic is a real or perceived imbalance of power between the bully and the target (Leymann, 1996; Niedl, 1995). That is, the target perceives that he or she is incapable of neutralizing or stopping the negative behaviours and that one feels powerless in the situation compared to the perpetrator. Typically, bullying develops as an escalating process where the person confronted ends up in an inferior position and becomes the target of systematic negative social acts (Einarsen *et al.*, 2003b). Hence, compared to other forms of negative and harassing treatment in the workplace, such as psychological aggression (e.g., Barling, Dupré, & Kelloway, 2009; Schat & Kelloway, 2004), incivility (e.g., Cortina, Magley, Williams, & Langhout, 2001), and sexual harassment (e.g., Fitzgerald, Swan, & Magley, 1997; Willness, Steel, & Lee, 2007), all which may take the form of single episodes, workplace bullying is characterized by being a long-lasting phenomenon with repeated episodes where the target have difficulties defending him or herself against behaviours that take place.

As is commonly the case in new areas of research, researchers within the field have struggled to establish a single agreed upon operationalization of workplace bullying. Yet, several studies on the phenomenology, antecedents, and outcomes of bullying have been conducted (cf. Hoel & Salin, 2003; Neuman & Baron, 2003; Zapf *et al.*, 2003). Because of this focus on nature and phenomenology, Keashly and Harvey (2005) claim that construct research on workplace bullying has generally been neglected because of a desire to discover substantive issues. That is, during more than 20 years of research, very few studies have critically investigated the use of methodology in research on workplace bullying. For instance, as there has been insufficient research on the development of psychometrically sound measures, the measurement of the phenomenon has not been as rigorous as one would hope (Keashly & Harvey, 2005). Hence, what we so far know about workplace bullying may actually be influenced by deficient measurement methods and research designs. Basically, research on workplace bullying seems to be biased by (a) variations in operational definitions and (b) problems with generalizability (cf. Agervold, 2007; Nielsen *et al.*, 2009; Schat *et al.*, 2006). In the upcoming sections, we will argue for why these issues are especially important to take into consideration when investigating the occurrence of workplace bullying.

Operationalizations of workplace bullying

Reviews of empirical findings on workplace bullying shows that a number of approaches and instruments of varying complexity have been devised to measure the phenomenon (Agervold, 2007; Cowie, Naylor, Rivers, Smith, & Pereira, 2002; Zapf *et al.*, 2003). Yet, most approaches have either assessed (a) the respondents' overall perception of being victimized by bullying (e.g., Einarsen & Skogstad, 1996) or (b) the respondents' perception of being exposed to a range of specific bullying behaviour (e.g., Salin, 2001; see also Zapf *et al.*, 2003). Examples of the two methods are displayed in Table 1.

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

Table 1. Examples of the self-labelling method and the behavioural experience method

Self-labelling method as presented by Einarsen and Skogstad (1996, pp. 190–191)

'*Bullying takes place when one or more persons systematically and over time feel that they have been subjected to negative treatment on the part of one or more persons, in a situation in which the person(s) exposed to the treatment have difficulty in defending themselves against them. It is not bullying when two equally strong opponents are in conflict with each other*'

According to this definition, have you been subjected to bullying at the workplace during the last 6 months

1. No
2. Yes, once or twice
3. Yes, now and then
4. Yes, about once a week
5. Yes, many times a week

Behavioural experience method. Sample items from the NAQ – Revised (Einarsen *et al.*, 2009)

The following behaviours are often seen as examples of negative behaviour in the workplace. Over the last 6 months, how often have you been subjected to the following negative acts at work?

Please circle the number that best corresponds with your experience over the last 6 months

	Never	Now and then	Monthly	Weekly	Daily
1. Being humiliated or ridiculed in connection with your work	1	2	3	4	5
2. Having key areas of responsibility removed or replaced with more trivial or unpleasant tasks	1	2	3	4	5
3. Persistent criticism of your work and effort	1	2	3	4	5
4. Being the subject of excessive teasing and sarcasm	1	2	3	4	5

When assessing respondents' perception of being victimized, often referred to as the self-labelling approach, participants are usually asked a single-item question about whether or not they have been bullied. In some studies, respondents are presented with a theoretical definition of bullying prior of being asked whether or not they, using the definition as a basis, would characterize their experiences in the workplace over a given period of time as bullying (e.g., Einarsen & Skogstad, 1996; O'Moore, Lynch, & Niamh, 2003). In other studies, the question about bullying has been asked without a preceding definition (e.g., Lewis, 1999; Rayner, 1997).

When investigating exposure to specific bullying behaviours, i.e., 'the behavioural experience method', respondents are presented with an inventory that includes various types of behaviour that may be named bullying if they occur on a repeated basis. The respondents are then asked to report how frequently they have experienced the different behaviours within a given time period. Commonly used inventories are the Leymann Inventory of Psychological Terror (LIPT; Leymann, 1990), the Negative Acts Questionnaire (NAQ; Einarsen, Hoel, & Notelaers, 2009; Einarsen & Raknes, 1997), and the Workplace Aggression Research Questionnaire (Harvey & Keashly, 2003). In studies using the behavioural experience approach, the prevalence of bullying is usually determined by a predefined operational criterion (e.g., Niedhammer, David, & Degioanni, 2006; Salin, 2001) or a statistical cut-off criterion (e.g., Björkqvist, Österman, & Hjeltbäck, 1994; Nielsen *et al.*, 2009; Ólafsson &

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

Jóhannsdóttir, 2004). With regard to the former, Leymann (1996) proposed a criterion of exposure to at least one negative act per week over a period of at least 6 months, whereas Mikkelsen and Einarsen (2001) claim that two negative acts are required to classify the experience as bullying.

The distinction between the self-labelling approach and the behavioural experience approach has important theoretical and methodological implications. Directly querying respondents about their self-labelled victimization from bullying, measures a perceptual construct; thus, reported prevalence rates will be influenced by individual characteristics of the respondent and, possibly, also by the particular psychological climate in the organization (Ilies, Hauserman, Schwochau, & Stibal, 2003). Magley, Hulin, Fitzgerald, and DeNardo (1999) describe a range of factors that may influence whether respondents will label their experiences as bullying or harassment. Firstly, in cases where the respondents are not given a definition of bullying, some may not self-label simply because the particular situation they experience does not conform to their personal definition of, and experiences with, bullying. Psychological defence mechanisms may be another problem, as some people may not want to see themselves as victims of bullying and therefore avoid labelling themselves as such. According to Magley *et al.* (1999), the excess meaning attached to the word victim may be integral to reluctance to label experiences as bullying. Because many people have negative views of victims, admitting victimization may be perceived as synonymous with weakness and inadequate coping, and may also imply some sort of stigmatization (Agervold, 2007; Lewis, 2004; O'Moore & Hillery, 1989). Hence, self-labelling as a victim of bullying may be a significant threat to self-esteem (van Beest & Williams, 2006), and the assessment by self-labelling may in fact produce an underestimation of the problem (Einarsen & Skogstad, 1996). Although some of the problems described above also apply to the behavioural experience approach, it is reasonable to assume that the self-protection issue may not have the same impact as the latter does not require the respondents to label themselves as victims. It has even been argued that when the behavioural experience approach is utilized, respondents may be more comfortable in reporting, and may even over-report their negative experience (Kokubun, 2007).

As for other methodological artifacts that may influence prevalence rates of the behavioural experience method, there are reasons to discuss the construct validity of the approach. As aggregation of observations seldom includes all aspects of a given construct, constructs are only approximately measured by summarizing different observations of the phenomenon in question (Nunnally & Bernstein, 1994). For example, when using popular inventories, such as the NAQ and the LIPT, a key element of the theoretical definition of bullying is omitted from the measure as the power imbalance between victim and perpetrator is not explicitly taken into consideration in the items of the scales. This method only assesses the frequency and duration of different bullying behaviour, whereas the power difference between target and perpetrator is not measured. Hence, we do not know whether the respondent can defend him or herself against the behaviour. As a consequence, the behavioural experience method may produce higher estimates than the self-labelling method because it does not take the perceived power difference between the parties into consideration.

Low construct validity may also be a problem when employing the self-labelling without definition approach. When using this method, it is somewhat uncertain to what extent the respondents' understanding of bullying actually corresponds with the

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

researcher's formal definition. For instance, in a comparison of formal definitions and lay definitions of workplace bullying, it was found that the respondents' lay definitions excluded central elements found in formal definitions, while they at the same time included elements that were not part of the formal definitions (Saunders, Huynh, & Goodman-Delahunty, 2007). While most lay persons included perpetration of negative behaviour as a defining characteristic, only 15.2% of the participants included power imbalance and only 14.7% included persistency. Furthermore, many lay definitions included themes of fairness and respect not found in the formal definition. Hence, by using the self-labelling without definition method, bullying is measured in very broad terms, with the risk of overestimating the prevalence rates as the respondents may report incidences that do not qualify as bullying according to researchers' understanding of the phenomenon.

In conclusion, a substantive question of interest with regard to the measurement of workplace bullying is whether surveys based on self-labelling with or without a given definition and behavioural experience surveys lead to different reported prevalence rates. Furthermore, if the different approaches indeed yield different results, what is the magnitude of the difference? Consequently, related to the aims of this study, we will provide cumulative estimates of the actual prevalence of workplace bullying, as measured by both self-labelling and behavioural experiences, and investigate differences between the approaches. Based on the above reasoning, we will explore two hypotheses about the relation between measurement methods and prevalence rates:

> *Hypothesis 1:* Studies using the behavioural experience method will produce higher prevalence rates of workplace bullying than studies based on self-labelling methods.
>
> *Hypothesis 2:* When using self-labelling methods to estimate prevalence of bullying, estimates based on an operational definition will give a lower prevalence rate than estimates where the bullying concept is not predefined.

Sampling in research on workplace bullying

A major challenge when collecting survey data is to achieve the best possible representation of the total population, while striking a balance between costs and efficiency on the one hand, and external validity, i.e., the ability to generalize the findings, on the other (cf. Smith, Hannaford, Elliot, Smith, & Chambers, 2005). The prevalence of workplace bullying has been investigated in a range of different samples. Some studies have utilized large nationwide representative samples (e.g., Niedhammer *et al.*, 2006; Nielsen *et al.*, 2009; O'Connell & Williams, 2002), whereas others are based on small convenience samples from individual enterprises or organizations (e.g., Thomas, 2004; Varhama & Björkqvist, 2004a).

Although non-random sampling has the advantage that the investigator obtains easy access to research participants without spending a great deal of time or money (Cozby, 2001), the external validity of such samples may be low, and the data highly skewed (Mullen, Pathé, Purcell, & Stuart, 1999). For example, Nielsen and Einarsen (2008) found a non-random sample of targets of bullying to differ from targets in a random and representative sample of the general workforce with the non-randomly sampled targets reporting significantly more frequent and long lasting exposure to bullying. The two samples also differed significantly on a range of demographical variables such as occupation, education, and age distribution.

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

The low external validity of non-random samples with regard to prevalence rates may be explained by various sample-selection biases. Ilies *et al.* (2003) describe two such biases. First, non-random sampling techniques may be biased towards samples that are likely to report higher prevalence rates through an availability mechanism. For instance, targets that have filed grievances or sought help are simply more available to researchers, and may therefore contribute to exaggerate the prevalence rates of bullying. However, Ilies and colleagues further argue that non-random samples may also bias the prevalence rates downwardly. Organizations with a heightened awareness about harassment and aggression are more likely to monitor bullying and cooperate with academic researchers in studying the phenomenon. Therefore, samples from such organizations may be more readily available for bullying researchers, and these samples are likely to report lower levels of workplace bullying.

In sum, the use of non-random samples may result in non-generalizable and non-valid findings, as well as biased responses that influence prevalence rates (cf. Arvey & Cavanaugh, 1995; Ilies *et al.*, 2003; O'Donohue, Downs, & Yeater, 1998). Since many studies on workplace bullying are based on non-random samples, it is important to empirically investigate the impact and magnitude of the type of sample on the reported prevalence rates. Hence, in this study, we will examine the impact of random and non-random sampling on prevalence rates of bullying. Although, from a theoretical perspective, it is possible that non-random samples may lead to both higher and lower prevalence rates than random samples, prior empirical findings indicate that non-random sampling leads to higher levels of bullying and harassment when comparing with random sampling techniques (cf. Ilies *et al.*, 2003; Nielsen & Einarsen, 2008). Hence, the arguments for lower prevalence in non-random samples seem to be out-weighted by the arguments for higher prevalence rates. Consequently, the following hypothesis will be tested:

> *Hypothesis 3:* Studies using non-random samples will produce higher prevalence rates of workplace bullying than studies based on random samples.

When conducting meta-analyses with studies from different countries there is always the risk of geographical biases. For instance, as a large part of studies on workplace bullying have their origin in the Scandinavian countries (Einarsen *et al.*, 2003b), the prevalence rates from these countries may have an impact on the overall rates. Hence, to investigate whether the occurrence of bullying is influenced by such bias, we also control for geographical differences in prevalence rates.

Method

Although meta-analysis has mostly been applied to correlations or effect sizes, the approach can be extended to cumulate proportions and rates (Borenstein, Hedges, Higgins, & Rothstein, 2009; Ilies *et al.*, 2003; Viswesvaran & Schmidt, 1992). Using a study design in line with Ilies *et al.* (2003) study of incidence rates of sexual harassment, we conducted an overall meta-analysis to cumulate the results from independent samples. Thereafter, multiple separate meta-analyses were performed to investigate the prevalence rates of bullying as delimited by the moderator variables.

To investigate cumulative prevalence rates of workplace bullying and the effects of methodological moderators, we based our meta-analysis on the procedure proposed by Hedges and Olkin (1985). When using this approach, meta-analytic evidence for the presence of moderators requires that estimates are different in the categories formed by

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

the potential moderator variable, and that these estimates are significantly heterogeneous (Borenstein, Hedges, Higgins, & Rothstein, 2009; Lipsey & Wilson, 2001). The degree of variability of the estimates across studies is assessed with the homogeneity statistic Q. In testing for moderating effects, each study is assigned a numerical value based in the moderator (e.g., 1 = random sample, 2 = non-random sample) and grouped according to the coding scheme. The difference between mean group prevalence rates is assessed by computing a between-group homogeneity statistic Q_B. The presence of a moderator is indicated by a statistically significant Q_B, which suggests a difference between the mean prevalence rate estimates across groups (cf. Aguinis, Struman, & Pierce, 2008). All meta-analyses were carried out with the comprehensive meta-analysis (version 2) software developed by Biostat (Borenstein, Hedges, Higgins, & Rothstein, 2005).

Random effects models are emphasized in this paper, although fixed effects findings are also reported. Random effects models, as opposed to regular fixed effects models, should be employed when accumulating data from a series of studies where it is assumed that the effect size varies from one study to the next, and where it is unlikely that studies are functionally equivalent (Borenstein, Hedges, & Rothstein, 2007; Lipsey & Wilson, 2001; Schulze, 2004). In the case of this meta-analysis, it is reasonable to assume that the studies included are not functionally equivalent as they vary with regard to factors such as country of origin, sample composition, and many other variables.

Literature search

To identify relevant studies, we followed the literature search strategies proposed by Durlak and Lipsey (1991). 'Bullying', 'mobbing', 'emotional abuse', 'verbal abuse', 'harassment', 'aggression', 'interpersonal conflict', and 'victimisation/victimization', all concepts which have been used to describe exposure to long-lasting and systematic harassment at work, were each combined with the words 'work', 'workplace', 'worksite', and 'organisation/organization' and entered into the PsychINFO, ISI Web of Science, Cochrane Library, Science Direct, Pubmed, Medlib, Norart, and Proquest databases. Internet searches via www.google.com and Google Scholar were performed to find other available literature. Further, the authors' personal collection of literature, gathered from about 1988 to September 2009, were examined to find missing publications. Citations in the gathered publications were inspected. As a last step, prominent authors within the field, and authors of papers that did not report prevalence, were contacted to obtain information on unpublished results. The study coding form followed the guidelines presented by Lipsey and Wilson (2001).

Studies included

To be included in this meta-analysis, studies had to report prevalence of a phenomenon in line with the definition of workplace bullying described in the introductory section of this paper. Hence, unlike prior meta-analyses on interpersonal aggression and harassment which have synthesized many different forms of mistreatment (e.g., Bowling & Beehr, 2006; Hershcovis *et al.*, 2007; Lapierre *et al.*, 2005), this meta-analysis focus exclusively on workplace bullying. That is, the exposure to persistent and systematic negative behaviours at the workplace. As a consequence, studies reporting one-off offences or sporadically incidences of aggression and harassment were excluded (e.g., Cortina *et al.*, 2001; Schat *et al.*, 2006).

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

The reported estimates were divided into categories based on the moderating variables. As a number of studies used more than one estimation methods within the same sample, some estimates are statistically dependent on each other. The studies were therefore classified according to utilized estimation method before sampling procedure was taken into consideration. To be included in the self-labelling categories, studies had to apply a question about victimization from bullying, either with or without a preceding operational definition. All studies that reported some sort of self-labelled victimization from bullying were included in the meta-analysis. Hence, this was used as inclusion criterion for self-labelling studies.

As for the behavioural approach, four main cut-off criteria for the occurrence of bullying have been employed in the literature. To have the most consistent measure possible, we only included behavioural studies employing the Leymann (1993) criteria of exposure to at least one negative act per week with duration of at least 6 month. The three other criteria, that is, the Mikkelsen and Einarsen (2001) criterion of exposure to at least two negative acts per week, the use of statistically calculated cut-off points and latent class cluster analysis, were excluded from the present meta-analysis for three reasons. First of all, these methods usually give very different estimates from the Leymann criterion (cf. Mikkelsen & Einarsen, 2001; Nielsen *et al.*, 2009; Notelaers, Einarsen, De Witte, & Vermunt, 2006), and the inclusion of these will therefore give a less uniform measure of behavioural experiences. Secondly, because these estimation methods have only been used in a limited number of studies, they cannot be made into a separate moderator category. Finally, most of these studies did also report findings comparable with the Leymann criterion. Hence, to avoid statistical dependency, only one behavioural experience estimate is included from each study.

With regard to sampling technique, we used a criteria and coding scheme which was developed by Ilies *et al.* (2003). To be included in the random-sample category, an estimate had to be based on (a) a national probability sample (random or stratified random) or (b) a random sample across multiple organizations or in a multiple-site organization (e.g., government or state employees), or (c) a sample that resulted from the sampling of the entire sampling frame (as defined by the study) in a single-site organization.

Of a total of several hundred studies on phenomena which could be classified as workplace bullying, 108 reported findings on prevalence, and thus met our first criterion. Twelve of these studies were excluded as they reported findings from a sample similar to another study (e.g., Hoel, 2002; Notelaers, Einarsen, *et al.*, 2006). Five studies investigated bullying in a setting that could not be reckoned as white or blue collar work, i.e., bullying among prison inmates (e.g., Ireland, 2004; Palmer & Thakordas, 2005; Power, Dyson, & Wozniak, 1997) and basic training army recruits (Østvik & Rudmin, 2001) and were therefore excluded from the meta-analysis. Nine studies were excluded because they reported exposure to single behaviour without further specification of duration and frequency (e.g., Quine, 1999; Richman, Flaherty, & Johnson, 1999; Schat *et al.*, 2006), while two studies were excluded because the assessment tool was not adequately described (Cox, 1987; Voss, Floderus, & Diderichsen, 2001). Of the included studies, 12 reported two estimates of prevalence rates based on the same sample (e.g., Mikkelsen & Einarsen, 2001; Niedhammer *et al.*, 2006; Salin, 2001), but with different types of estimation methods (self-labelling vs. behavioural experience). In such cases, both estimates were included in the study, but placed in different categories to avoid statistical dependency. Hence, altogether 102 estimates of prevalence of workplace bullying from 86 different samples were included

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

in the analysis. Total sample size for the 86 samples is 130,973 respondents. All studies included in the meta-analysis are marked with an asterisk in the reference list.

Results

Description of the studies

Of 102 different estimates of bullying, 34 were based on the behavioural experience method, 21 were based on the self-labelling without definition method, whereas 47 were based on the self-labelling with definition method. Most of the 34 behavioural experience studies utilized either a variation of the NAQ (47%) or a variation of the LIPT (32%). Reporting period varied from point prevalence to lifetime prevalence, although the majority of the studies reported 6–12 months prevalence. With regard to sampling procedure, 48% of the estimates were based on a random sample. Mean response rate for non-random samples was 57.4%, whereas the mean rate for random samples was 56.94%. Hence, the average response rates are somewhat higher than mean rate for organizational research in general (cf. Baruch & Holtom, 2008). The difference in response rate between the random and the non-random samples was non-significant.

Samples from 24 different countries and one multinational sample (Jennifer, Cowie, & Ananiadou, 2003) were represented in the meta-analysis. The majority (68%) of the countries are situated in Europe. Yet, every settled continent was represented with at least one study. A clear bias towards the Scandinavian countries prevailed, as more than one-fourth of the included samples came from Denmark (10), Norway (10), or Sweden (2). The majority (81.8%) of the studies included was published in the period 2000–2008, whereas 15.6% was published in the 1990s. Only two studies (2.6%) were published in the 1980s.

Prevalence rate estimates and the impact of methodological moderators

Analyses were conducted in two steps. In the first step, we investigated the direct effects of each potential moderator variable on prevalence rates. In the second step, we conducted separate meta-analyses for each of the categories of variables in order to estimate the prevalence rate of workplace bullying in the categories delimited by moderator variables. An overall rate, weighted by sample size, of .136 was found across all estimates. As this overall rate contains both self-labelling and behavioural experience estimates, where some are derived from the same sample, this number is not meaningful by itself since the estimates from the identical samples are statistically dependent (cf. Hunter & Schmidt, 2004; Ilies *et al.*, 2003). When removing the statistically dependent samples, the total K was reduced to 70 studies with a total of 112,714 respondents. For these studies, an overall rate of .146 (95% CI [.123, .172]; $K = 70$) was established.

When investigating measurement method as a moderator on the prevalence rates, the findings showed an average rate of .148 (95% CI [.119, .182]; $K = 34$) for behavioural experiences, .181 (95% CI [.129, .248]; $K = 21$) for self-labelling without definition, and .113 (95% CI [.093, .137]; $K = 47$) for self-labelling with definition. The estimates were found to be significantly heterogeneous ($Q_B = 6.85$, $df = 2$, $p < .05$).

To investigate differences between random and non-random samples, statistically dependent samples, i.e., studies using different measurement methods based on an identical sample, were removed from analysis to achieve an unbiased average rate. A statistically significant difference ($Q_B = 10.88$, $df = 1$, $p < .001$) was found between

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

random (rate = .113; 95% CI [.094, .136]; $K = 39$) and non-random (rate = .200; 95% CI [.151, .259]; $K = 31$) sampled studies.

Differences between random and non-random samples were also investigated within each of the measurement method categories. Weighted prevalence rates, 95% CI, and Q values from these analyses are found in Table 2. The findings show that no significant differences were found between random (.144) and non-random (.145) sampled ($Q_B = 0.00$, $df = 1$, $p > .05$) behavioural estimates. Similar results were obtained for studies using self-labelling without definition ($Q_B = 0.17$, $df = 1$, $p > .05$). However, for studies using the self-labelling with definition method, the findings showed significant differences between random (.093) and non-random (.155) samples ($Q_B = 8.42$, $df = 1$, $p < .01$), thus indicating heterogeneity.

Table 2. Meta-analysis of reported prevalence rates of workplace bullying: methodological moderators

	Size		Random effects model			Fixed effects model		
Moderators	*K*	*N*	Mean rate	95% CI	Q_B	Mean rate	95% CI	Q_B
Behavioural experiences	34	44,477	.148	.119–.182	6.84*	.173	.169–.177	1,091.99**
Random	15	27,860	.144	.114–.181	0.00^{ns}	.172	.168–.177	0.40^{ns}
Non-random	19	16,617	.145	.092–.222		.175	.168–.182	
Self-labelling without definition	21	15,103	.181	.129–248		.227	.219–.234	
Random	6	5,974	.174	.150–.201	0.17^{ns}	.182	.172–.192	123.69**
Non-random	15	9,129	.195	.116–.309		.270	.258–.282	
Self-labelling with definition	47	89,652	.113	.093–.137		.122	.120–.125	
Random	28	75,607	.093	.074–.116	8.42**	.097	.094–.099	1,441.19**
Non-random	19	14,045	.155	.119–.199		.218	.211–.225	

95% CI: 95% Confidence Interval; *Difference significant at $p < .05$; **difference significant at $p < .01$.

When using fixed effects models to examine the impact of the moderators on prevalence rates of bullying, significant differences were found between all moderator categories except for sampling procedure within the behavioural experience study category. Hence, these findings mainly supported our hypotheses. Prevalence rates, 95% CI, and Q_B-statistics for the fixed effects models are displayed in Table 2.

Geographical differences

To investigate whether the above findings are influenced by geographical differences, the included studies were divided into three groups based on country of origin. The first group comprised the Scandinavian countries, whereas the second group included other European countries. Because of low K in other regions, the non-European countries were categorized into one single group. Two sets of analyses were conducted. In the first set, prevalence rates were compared across the three categories of countries. In the second set, subanalyses on the impact methodological moderators were conducted within the categories. Table 3 displays the fixed and random effects average rates for the different measurement methods separated by the three groups of countries.

When comparing findings on measurement method across countries, significant differences were established for all measurement methods. As shown for the random

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

Table 3. Average prevalence rates for measurement method in different regions

			Random effects model		Fixed effects model	
Countries	Measurement method	*K*	Mean rate	95% CV	Mean rate	95% CV
Scandinavia	Behavioural experiences	12	.095	.064–.139	.105	.099–.112
	Self-labelling without definition	5	.065	.025–.158	.057	.049–.065
	Self-labelling with definition	13	.046	.035–.061	.063	0.60–.066
Other European countries	Behavioural experiences	14	.157	.124–.196	.171	.167–.176
	Self-labelling without definition	9	.243	.156–.357	.256	.245–.267
	Self-labelling with definition	24	.138	.108–.174	.140	.137–.143
Non-European countries	Behavioural experiences	8	.257	.148–.408	.321	.305–.337
	Self-labelling without definition	6	.277	.191–.384	.255	.241–.271
	Self-labelling with definition	10	.198	.129–.291	.214	.200–.229

effects in Table 3, the Scandinavian studies reported lower rates for behavioural experience estimates ($Q_B = 9.28$, $df = 1$, $p < .01$), self-labelling without definition ($Q_B = 9.32$, $df = 1, p < .05$), and self-labelling with definition ($Q_B = 47.53$, $df = 1, p < .001$). As for sampling method, significant heterogeneity was found for random sampled studies ($Q_B = 33.13$, $df = 1$, $p < .001$). The findings show that the lowest rate was found for the Scandinavian countries. For non-random sampled studies, the countries were found to be homogeneous ($Q_B = 0.87$, $df = 1$, $p > .05$). However, this latter finding may be explained by the low number of non-random sampled studies from Scandinavia.

In the second set of analyses, the impact of the methodological moderators was investigated within the groups of countries. For the Scandinavian countries, significant differences were found between the measurement methods ($Q_B = 8.7$, $df = 2$, $p < .01$). As displayed in Table 3, the lowest rates were found for self-labelling with definition and self-labelling without definition studies, whereas behavioural experience studies produced the highest rate. Because of a high degree of statistical dependency between the studies, it was not possible to conduct comparisons between sampling methods in the Scandinavian subgroup. In the 'other European countries' group, the different measurement methods were found to provide homogeneous estimates ($Q_B = 4.87$, $df = 2$, $p > .05$). Nonetheless, the lowest rate was found for self-labelling with definition studies, whereas self-labelling without definition studies produced the highest rates (cf. Table 3), thus confirming the pattern found in the overall analyses. Homogeneous estimates were also revealed with regard to sampling technique ($Q_B = 1.76$, $df = 1$, $p > .05$). The analyses for the non-European countries provided similar results as for the 'other European countries' with regard to measurement method ($Q_B = 1.59$, $df = 2$, $p > .05$). Because of low K for random sampled studies, no comparisons were made with regard to sampling procedure.

Discussion

The main objectives of this study were to provide cumulative estimates of the prevalence of workplace bullying, and to investigate the influence of methodological artifacts on reported prevalence rates. Across the studies included, an overall estimate of

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

14.6% was established. Yet, such an overall estimate must be interpreted with caution as the further findings confirm our hypotheses that prevalence rates of workplace bullying are influenced by measurement methods and sampling procedures. With an overall rate of about 11%, the results show that studies using the self-labelling with definition method provide the lowest estimate of bullying. When using the behavioural experience method, the prevalence is somewhat higher with rates about 15%, whereas the self-labelling without definition method yields the highest estimates with overall rates of 18%. Hence, compared to studies using self-labelling with definition, the prevalence rates of this latter approach are almost twice as high. With regard to sampling technique, the findings show that non-random samples at an average yield 8.7% point higher estimate of workplace bullying than do random samples. Yet, when examining sampling technique within the measurement method categories, sampling method had less impact on the prevalence estimates.

Based on the present meta-analysis, workplace bullying seems to be a substantial problem in contemporary working life. At a worldwide average, the findings show that at least 1 out of 10, and maybe as many as about 1 out of 5, workers are exposed to bullying in their workplaces. As bullying and harassment have been shown to have considerable negative effects both for organizations and those exposed (Bowling & Beehr, 2006; Einarsen & Mikkelsen, 2003; Hoel, Einarsen, & Cooper, 2003), our findings demonstrate that workplace bullying is a problem that should be taken into consideration in organizational life. Hence, governments, employers, and organizations need to take the bullying issue seriously by developing and implementing interventions and strategies to handle the problem.

Compared to adjacent phenomena such as sexual harassment and aggression, this study shows that workplace bullying has lower prevalence rates. For instance, in a meta-analysis of the incidence of sexual harassment, Ilies *et al.* (2003) found that 58% reported exposure to sexual harassing behaviour, whereas 24% self-labelled as victims of such harassment. In a representative survey of aggression among US workers, it was found that 41.4% had experienced at least one instance of psychological aggression at work during the past 12 months (Schat *et al.*, 2006). Although based on inconsistent reporting periods, the lower prevalence rates found in our meta-analysis support the assumption that workplace bullying is a phenomenon distinct from other forms of interpersonal aggression. That is, whereas aggression and harassment may be described by single and isolated behaviours, workplace bullying is per definition based on repeated behaviours over a prolonged time period. Consequently, because more frequent negative acts are needed to describe the bullying phenomenon, prevalence rates should also be lower compared to aggression and harassment.

Disparities between estimation methods

As there is little agreement between the different estimation methods with regard to prevalence rates, two questions come to mind. Firstly, since the methods aim at measuring the same phenomenon, why do they produce different prevalence rates? Secondly, which estimate is most trustworthy? Answering these two questions can have important implications for future research, because knowledge about strengths and problems associated with the measurements may serve as guidelines for which method to use when measuring workplace bullying in future studies. Reliable and valid findings may also have important implications for practice, as initiatives and endeavours to

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

handle bullying are dependent on knowledge about the frequency and nature of the phenomenon (cf. O'Donohue *et al.*, 1998).

The finding of rate disparities between the estimation methods may have many possible origins. One explanation could be that the degree of measurement accuracy varies. For instance, as the behavioural experience method investigates exposure to a range of different negative acts, it may be perceived as more accurate than the self-labelling method since this latter approach is more dependent on the respondent's understanding of the bullying phenomenon. This may especially hold true when a definition is not offered.

Another explanation for the extensive rate disparity between the estimates may be that different aspects of the bullying phenomenon are measured with the methods. An important difference between the two self-labelling approaches and the behavioural experience approach is that while the former ask whether the respondent perceives her or himself as a being bullied or not, this is not a criterion when using the behavioural experience approach. This latter method solely relies on the respondent's reported exposure to specific unwanted behaviours, and it is therefore up to the researcher to define who is bullied, and who is not. Our claim is therefore that the two approaches should be compared with care, as they seem to assess different facets of workplace bullying.

Nevertheless, as long as the self-labelling method and the behavioural experience method capture different features of workplace bullying, it can also be argued that both measures provide valid, but supplementary, information on the phenomenon (Nielsen *et al.*, 2009). Future studies on bullying should therefore combine the methods in order to both measure respondents' exposure to persistent bullying behaviour and their perception of being victimized by this behaviour (cf. Mikkelsen & Einarsen, 2001; Salin, 2001). By doing so, one will gain information about both the nature and frequency of the behaviour, and about whether or not the respondents experience the exposure to these behaviours as bullying.

The remarkable rate disparity between the two different self-labelling methods found in this meta-analysis suggest that it is not without consequence how victimization from bullying is assessed. As the occurrence of bullying is significantly lower when using the self-labelling with definition approach, the findings indicate that theoretical definitions of bullying are much stricter than lay definitions. As discussed in the introductory section of this paper, one increases the risk of potential misperceptions when an exact and precise definition of bullying is absent. That is, it is quite uncertain to what degree the respondents understanding of the bullying concept corresponds with the researcher's formal definition. Because of the large difference found between self-labelling with definition and self-labelling without definition found in this study, our view is that the former should be employed in future studies because this method seems to provide the best operationalization of the theoretical definition of bullying when using the self-labelling approach.

Sampling respondents

With regard to how sampling method influences prevalence rates, the findings confirm our hypothesis that non-random samples provide the highest rates. The results show an 8.7% point higher rate in non-random samples compared to random samples. As discussed earlier in this article, there may be several reasons for why non-random samples yield higher rates than random samples. In addition to the presented

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

explanations, response rates have been proposed as a factor influencing survey data (cf. Baruch, 1999; Baruch & Holtom, 2008). In the present study, equal response rates were found for random and non-random samples, thus indicating that non-response level is the same for both sampling techniques. However, it may still be possible that there are differences among the non-response respondents in random and non-random samples. For instance, as non-random sampling surveys may have a more specific focus on the workplace bullying problem than random sampled studies (cf. Nielsen & Einarsen, 2008), it could be that the former has the problem that people who have nothing to do with bullying more often assume that they do no need to respond to the survey because they cannot contribute to the problem anyway.

Assuming that the rate found for random samples can be used as a valid baseline for comparison, the findings of this study indicate that non-random sampling increases the risk of overestimating prevalence rates. Thus, the results support the assumption about limited or insignificant external validity of non-random samples (Cozby, 2001). In other words, findings from non-random samples may not necessarily be generalized beyond the samples at hand, and their utility for providing information concerning the wider prevalence of workplace bullying is questionable.

However, the use of non-random samples should not be rejected in all settings. Non-random samples are generally cost effective and may be quite useful for investigating tendencies and phenomenological aspects of workplace bullying when representative prevalence rates are not a main objective. Thus, it is important to distinguish between representative studies that aim to assess the frequency of workplace bullying in the general population on the one hand, and those that attempt to shed light on the phenomenology and nature of bullying on the other (cf. Cozby, 2001; Matthiesen & Einarsen, 2001). Moreover, as non-random convenience samples more easily can capture respondents outside working life (cf. Nielsen & Einarsen, 2008), non-random sampling can also be a valuable means of accessing subgroups of respondents who are otherwise difficult to access through general surveys aimed at the employed part of the workforce, e.g., persons on long-term sick leave, pensioners, and unemployed.

Geographical differences in prevalence rates

In addition to the importance of methodological moderators, the findings also show that geographical factors should be taken into consideration when comparing prevalence rates of workplace bullying. Although only a limited number of regions were included, the analyses indicate that the findings on prevalence rates may be dependent on geographical factors. In agreement with previous research (Nielsen *et al.*, 2009; Zapf *et al.*, 2003), a lower prevalence of bullying was found for the Scandinavian countries when comparing with other European countries and non-European countries. Hence, because of the overweight of Scandinavian studies included in this meta-analysis, the overall prevalence rates should be interpreted with care as they are probably influenced by the Scandinavian rates. Yet, it should be noted that the subanalysis within the geographical regions are somewhat restricted as some of the moderator categories only included a small number of studies. Furthermore, it should also be mentioned that the predominance of studies from the Scandinavian countries is most probably explained by the fact that a major part of research on workplace bullying in general stems from this particular geographical region (cf. Einarsen *et al.*, 2003b), and not because of a selection bias in the literature search.

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

A noteworthy finding in the Scandinavian countries category is that the self-labelling with and without definition provides very similar prevalence rates. Hence, presenting a definition of bullying seems to have less impact in Scandinavian than in other regions. A possible explanation for this finding is that the bullying concept has received much attention in Scandinavian schools and workplaces through research studies, interventions, and media coverage over a longer period of time than is the case in most other countries (e.g., Einarsen *et al.*, 2003b; Olweus, 1991). Consequently, because the concept is well-known and established among employees in Scandinavia, it may not need further clarification in survey studies (cf. Agervold & Mikkelsen, 2004).

Methodological strengths and limitations

Meta-analysis has been viewed as an efficient approach to synthesize research findings, especially since stronger conclusions may be reached, compared to traditional impressionistic literary reviews (cf. Hunter & Schmidt, 2004; Lipsey & Wilson, 2001). Hence, meta-analysis has the advantage of moving discussion away from the individual studies towards an overview of the whole body of research on a given topic. Nevertheless, meta-analyses are only as good as the studies included (Lipsey & Wilson, 2001), and cannot improve the quality or reporting of the original studies.

A commonly quoted bias in meta-analysis is the 'file drawer issue'. That is, many studies in a given field of research may be conducted but never published, and those that are not published, may well report different results from those that are included. For instance, studies with significant results are more likely to get published than are studies without significant results (Lipsey & Wilson, 2001). Among published studies, those with significant results are more likely to get published in English, more likely to be cited, and more likely to be cited repeatedly, leading to English language bias, citation bias, and multiple publication bias (Egger & Smith, 1998). To avoid the 'file drawer' problems and the language bias in the present study, we asked prominent researchers within the field about additional studies to those found in the major databases. Furthermore, as we have included studies published in a wide range of languages other than English, there is hardly any reason to assume a strong English language bias in our analyses.

Finally, it should be noted that there are, despite the cumulative strength of meta-analysis, several issues which a meta-analysis is not able to resolve and which should be considered when interpreting the findings. Reliance on self-report survey data is one commonly cited issue, raising concerns regarding socially desirable responses, as well as other data collection phenomena, such as demand characteristics (Willness *et al.*, 2007).

Conclusions and implications

This meta-study shows that prevalence rates of workplace bullying are dependent on methodological moderator variables, implying that research findings from different studies cannot be directly contrasted without taking moderators into account. For instance, one cannot explain the difference in prevalence rates in Denmark (Agervold, 2007) and Turkey (Bilgel *et al.*, 2006) with cultural differences, without first taking design issues into account. Furthermore, assuming that different aspects of the bullying phenomenon are measured with different methods, one should be careful with accumulating findings on workplace bullying without incorporating the methodological

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

characteristics of the studies included. Hence, to be able to make valid comparisons between studies, future research needs to agree upon a uniform measure of bullying, as well as a reliable and valid method for estimating the prevalence of the phenomenon.

The findings from this paper represent an ethical challenge for researchers, practitioners, and organizations. As the prevalence for bullying vary extensively depending upon the measurement method and sampling procedures being used, the research design can actually be used to manipulate the observed extent of the phenomenon in a given sample or enterprise. For instance, if a relatively low rate of bullying is wanted, the self-labelling with definition method can be chosen, whereas the self-labelling without definition could be used if a higher prevalence rate is 'desired', at least outside of the Scandinavian countries. Hence, researchers and practitioners need to be cautious when investigating bullying. However, workplace bullying is a complex phenomenon that seems to be difficult to assess using one single method. The most adequate solution for researchers and practitioners who want to investigate the extent of bullying without risking over or underestimating the phenomenon may therefore be to combine the self-labelling with the behavioural experience approach. Using such a combination, the former method will provide the number of employees who actually label themselves as victims of bullying, whereas the latter will provide information about the number of employees who are exposed to bullying behaviour.

As no valid baseline for the prevalence rates for bullying exists, it may be difficult for practitioners and researchers to know whether a revealed prevalence is high, medium, or low. The present study may therefore contribute to the field by presenting estimates of the global frequency of bullying, controlled for the effects of measurement method and sampling procedures. Finally, although bullying has been found to be a distinct phenomenon from many other forms of interpersonal aggression (Lapierre *et al.*, 2005; Parkins, Fishbein, & Ritchey, 2006), the behaviours which constitute workplace bullying, as well as the effects of these behaviours, are consistent with interpersonal harassment in general (Schat *et al.*, 2006; Walsh & Clarke, 2003). Hence, the findings of this study should also be of relevance to studies on other forms of interpersonal aggression.

Acknowledgements

The authors would like thank the following persons for their assistance with identifying and reading the Dutch, Spanish, Portuguese, and Finnish language papers included in this meta-analysis: Elfi Baillien, Miguel Martínez-Lugo, Guy Notelaers, Øyvind Nygaard Ringdal, and Denise Salin. We would also like to thank all authors who responded to our enquiries about their individual studies. Finally, we thank Prof. Evert Van de Vliert and members of The Bergen Bullying Research Group for helpful comments and suggestions.

References

References marked with an asterisk indicate studies included in the meta-analysis.

*Abe, K. (2007). *Hierarchical models of workplace bullying among Japanese hospital nurses*. Minneapolis, MN: University of Minnesota.

*Agervold, M. (2007). Bullying at work: A discussion of definitions and prevalence, based on an empirical study. *Scandinavian Journal of Psychology*, *48*, 161–172.

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

*Agervold, M. (2009). The significance of organizational factors for the incidence of bullying. *Scandinavian Journal of Psychology*, *50*(3), 267–276.

*Agervold, M., & Mikkelsen, E. G. (2004). Relationships between bullying, psychosocial work environment and individual stress reactions. *Work and Stress*, *18*(4), 336–351.

Aguinis, H., Struman, M. C., & Pierce, C. A. (2008). Comparison of three meta-analytic procedures for estimating moderating effects of categorical variables. *Organizational Research Methods*, *11*(1), 9–34.

*Arroyo, J., Martínez, M., & Pérez, J. (2004). *El acoso psicológico en el trabajo: Descripción del fenómeno en una muestra de empleados en Puerto Rico [Workplace mobbing: Phenomenon description in a sample of employees in Puerto Rico]*. Paper presented at the 51th Annual Convention of the Puerto Rico Psychology Association.

Arvey, R. D., & Cavanaugh, M. A. (1995). Using surveys to assess the prevalence of sexual harassment: Some methodological problems. *Journal of Social Issues*, *51*(1), 39–52.

*Ayers, K. M. S., Thomson, W. M., Newton, J. T., Morgaine, K. C., & Rich, A. M. (2009). Self-reported occupational health of general dental practitioners. *Occupational Medicine – Oxford*, *59*(3), 142–148.

*Baillien, E., Neyens, I., De Witte, H., & Vanoirbeek, K. (2005). *Ongewenst grensoverschrijdend gedrag op het werk: op welke manier speelt de organisatie een rol? Een kwantitatieve studie van risicofactoren op niveau van job, team en organisatie [Negative acts at work: How does the organization play a role. A quantitative study of task-, team- and organizational risk factors]*. Leuven: Departement Psychologie.

*Bairy, K. L., Thirumalaikolundusubramanian, P., Sivagnanam, G., Saraswathi, S., Sachidananada, A., & Shalini, A. (2007). Bullying among trainee doctors in southern India: A questionnaire study. *Journal of Postgraduate Medicine*, *53*(2), 87–91.

Barling, J., Dupré, K. E., & Kelloway, E. K. (2009). Predicting workplace aggression and violence. *Annual Review of Psychology*, *60*, 671–692.

Baruch, Y. (1999). Response rate in academic studies – a comparative analysis. *Human Relations*, *52*(4), 421–438.

*Baruch, Y. (2005). Bullying on the net: Adverse behavior on e-mail and its impact. *Information and Management*, *42*, 361–371.

Baruch, Y., & Holtom, B. C. (2008). Survey response rate levels and trends in organizational research. *Human Relations*, *61*(8), 1139–1160.

*Bilgel, N., Aytac, S., & Bayram, N. (2006). Bullying in Turkish white-collar workers. *Occupational Medicine*, *56*, 226–231.

Björkqvist, K., Österman, K., & Hjeltbäck, M. (1994). Aggression among university employees. *Aggressive Behavior*, *20*, 173–184.

Borenstein, M., Hedges, L. V., Higgins, J. P. T., & Rothstein, H. R. (2005). *Comprehensive meta-analysis version 2*. Englewood, NJ: Biostat.

Borenstein, M., Hedges, L. V., Higgins, J. P. T., & Rothstein, H. R. (2009). *Introduction to meta-analysis*. Chichester: Wiley.

Borenstein, M., Hedges, L. V., & Rothstein, H. R. (2007). *Meta-analysis. Fixed effects vs. random effects*. Englewood, NJ: Biostat.

Bowling, N. A., & Beehr, T. A. (2006). Workplace harassment from the victim's perspective: A theoretical model and meta analysis. *Journal of Applied Psychology*, *91*(5), 998–1012.

*Cemaloglu, N. (2007). The exposure of primary school teachers to bullying: An analysis of various variables. *Social Behavior and Personality*, *35*(6), 789–802.

Cortina, L. M., Magley, V. J., Williams, J. H., & Langhout, R. D. (2001). Incivility in the workplace: Incidence and impact. *Journal of Occupational Health Psychology*, *6*(1), 64–80.

Cowie, H., Naylor, P., Rivers, I., Smith, P. K., & Pereira, B. (2002). Measuring workplace bullying. *Aggression and Violent Behavior*, *7*, 33–51.

Cox, H. C. (1987). Verbal abuse in nursing: Report of a study. *Nursing Management*, *18*(11), 47–50.

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

*Coyne, I., Chong, P. S.-L., Seigne, E., & Randall, P. (2003). Self and peer nominations of bullying: An analysis of incident rates, individual differences, and perceptions of the working environment. *European Journal of Work and Organizational Psychology*, *12*(3), 209–228.

Cozby, P. C. (2001). *Methods in behavioral research* (7th ed.). Mountain View, CA: Mayfield Publishing.

*Daniels, D.-L. (2005). *Workplace bullying or bad behaviour in Australian organisations: Prevalence and employers' and employees' perceptions and interpretations of the phenomenon*. Unpublished Postgraduate Diploma of Psychology Thesis, Swinburne University of Technology, Swinburne.

Durlak, J. A., & Lipsey, M. W. (1991). A practitioner's guide to meta-analysis. *American Journal of Community Psychology*, *19*(3), 291–332.

Egger, M., & Smith, G. D. (1998). Meta-analysis. Bias in location and selection of studies. *British Medical Journal*, *316*, 61–66.

Einarsen, S. (2000). Harassment and bullying at work: A review of the Scandinavian approach. *Aggression and Violent Behavior*, *5*(4), 379–401.

Einarsen, S., Hoel, H., & Notelaers, G. (2009). Measuring bullying and harassment at work: Validity, factor structure, and psychometric properties of the Negative Acts Questionnaire – Revised. *Work and Stress*, *23*(1), 24–44.

Einarsen, S., Hoel, H., Zapf, D., & Cooper, C. L. (2003a). The concept of bullying at work: The European tradition. In S. Einarsen, H. Hoel, D. Zapf, & C. L. Cooper (Eds.), *Bullying and emotional abuse in the workplace. International perspectives in research and practice* (pp. 3–30). London: Taylor & Francis.

Einarsen, S., Hoel, H., Zapf, D., & Cooper, C. L. (Eds.), (2003b). *Bullying and emotional abuse in the workplace. International perspectives in research and practice*. London: Taylor & Francis.

Einarsen, S., Matthiesen, S. B., & Hauge, L. J. (2008). Bullying and harassment at work. In S. Cartwright & C. L. Cooper (Eds.), *The Oxford handbook of personnel psychology* (pp. 464–495). Oxford: Oxford University Press.

*Einarsen, S., Matthiesen, S. B., & Skogstad, A. (1998). Bullying, burnout and well-being among assistant nurses. *Journal of Occupational Health and Safety*, *14*(6), 563–568.

Einarsen, S., & Mikkelsen, E. G. (2003). Individual effects of exposure to bullying at work. In S. Einarsen, H. Hoel, D. Zapf, & C. L. Cooper (Eds.), *Bullying and emotional abuse in the workplace. International perspectives in research and practice* (pp. 127–144). London: Taylor & Francis.

*Einarsen, S., & Raknes, B. I. (1997). Harassment in the workplace and the victimization of men. *Violence and Victims*, *12*(3), 247–263.

*Einarsen, S., & Skogstad, A. (1996). Bullying at work: Epidemiological findings in public and private organizations. *European Journal of Work and Organizational Psychology*, *5*, 185–201.

*Eriksen, W., & Einarsen, S. (2004). Gender minority as a risk factor of exposure to bullying at work: The case of male assistant nurses. *European Journal of Work and Organizational Psychology*, *13*(4), 473–492.

Fitzgerald, L. F., Swan, S., & Magley, V. J. (1997). But was it really sexual harassment? Legal, behavioral, and psychological definitions of the workplace victimisation of women. In W. O'Donohue (Ed.), *Sexual harassment: Theory, research and treatment* (pp. 5–28). Boston, MA: Allyn & Bacon.

*Fox, S., & Stallworth, L. E. (2005). Racial/ethnic bullying: Exploring links between bullying and racism in the US workplace. *Journal of Vocational Behavior*, *66*, 438–456.

*Gil-Monte, P. R., Carretero, N., & Luciano, J. V. (2006). Prevalencia del mobbing en trabajadores de centros de asistencia a personas con discapacidad [Prevalence of mobbing in a sample of employees working with disabled people]. *Revista de Psicologia del Trabajo y de las Organizaciones*, *22*(3), 275–292.

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

*Hansen, Å. M., Høgh, A., Persson, A., Karlson, B., Garde, A. H., & Ørbæk, P. (2006). Bullying at work, health outcomes, and physiological stress response. *Journal of Psychosomatic Research*, *60*, 63–72.

Harvey, S., & Keashly, L. (2003). Predicting the risk for aggression in the workplace: Risk factors, self-esteem and time at work. *Social Behavior and Personality*, *31*(8), 807–814.

Hedges, L. V., & Olkin, I. (1985). *Statistical methods for meta-analysis*. Orlando, FL: Academic Press.

Hershcovis, M. S., & Barling, J. (2009). *Comparing the outcomes of sexual harassment and workplace aggression: A meta-analysis*. Unpublished paper.

Hershcovis, M. S., Turner, N., Barling, J., Arnold, K. A., Dupré, K. E., Inness, M., … Sivanathan, N. (2007). Predicting workplace aggression: A meta-analysis. *Journal of Applied Psychology*, *92*(1), 228–238.

Hoel, H. (2002). *Bullying at work in Great Britain*. Manchester: University of Manchester Institute of Science and Technology.

*Hoel, H., Cooper, C. L., & Faragher, B. (2001). The experience of bullying in Great Britain: The impact of organizational status. *European Journal of Work and Organizational Psychology*, *10*, 443–465.

Hoel, H., Einarsen, S., & Cooper, C. L. (2003). Organisational effects of bullying. In S. Einarsen, H. Hoel, D. Zapf, & C. L. Cooper (Eds.), *Bullying and emotional abuse in the workplace. International perspectives in research and practice* (pp. 145–162). London: Taylor & Francis.

Hoel, H., & Salin, D. (2003). Organisational antecedents of workplace bullying. In S. Einarsen, H. Hoel, D. Zapf, & C. L. Cooper (Eds.), *Bullying and emotional abuse in the workplace: International perspectives in research and practice* (pp. 203–218). London: Taylor & Francis.

*Høgh, A., & Dofradottir, A. (2001). Coping with bullying in the workplace. *European Journal of Work and Organizational Psychology*, *10*(4), 485–495.

Hunter, J. E., & Schmidt, F. L. (2004). *Methods of meta-analysis. Correcting error and bias in research findings* (2nd ed.). Thousand Oaks, CA: Sage.

Ilies, R., Hauserman, N., Schwochau, S., & Stibal, J. (2003). Reported incidence rates of work-related sexual harassment in the United States: Using meta-analysis to explain reported rate disparities. *Personnel Psychology*, *56*, 607–631.

Ireland, J. L. (2004). Nature, extent, and causes of bullying among personality-disordered patients in a high-secure hospital. *Aggressive Behavior*, *30*, 229–242.

*Jennifer, D., Cowie, H., & Ananiadou, K. (2003). Perceptions and experience of workplace bullying in five different working populations. *Aggressive Behavior*, *29*, 489–496.

*Justicia, F. J., Muñoz, J. L. B., & de Haro, E. F. (2006). Caracterización del acoso psicológico en el contexto universitario [Description of bullying in the university context]. *Revista de Psicologia del Trabajo y de las Organizaciones*, *22*(3), 293–308.

*Kaucsek, G., & Simon, P. (1995). *Psychoterror and risk-management in Hungary.* Paper presented at the 7th European Congress of Work and Organizational Psychology, Györ, Hungary.

Keashly, L., & Harvey, S. (2005). Emotional abuse in the workplace. In S. Fox & P. E. Spector (Eds.), *Counterproductive behavior. Investigations of actors and targets* (pp. 201–236). Washington, DC: American Psychological Association.

*Keashly, L., & Jagatic, K. (2000). *The nature, extent, and impact of emotional abuse in the workplace: Results of a statewide survey.* Paper presented at the Symposium on Persistent Patterns of Aggressive Behavior at Work, Toronto, ON.

Keashly, L., & Jagatic, K. (2003). By another name: American perspectives on workplace bullying. In S. Einarsen, H. Hoel, D. Zapf, & C. L. Cooper (Eds.), *Bullying and emotional abuse in the workplace. International perspectives in research and practice* (pp. 31–61). London: Taylor & Francis.

*Kivimäki, M., Elovainio, M., & Vathera, J. (2000). Workplace bullying and sickness absence in hospital staff. *Occupational and Environmental Medicine*, *57*, 656–660.

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

*Kivimäki, M., Leino-Arjas, P., Virtanen, M., Elovainio, M., Keltikangas-Järvinen, L., Puttonen, S., … Vahtera, J. (2004). Work stress and incidence of newly diagnosed fibromyalgia. Prospective cohort study. *Journal of Psychosomatic Research*, *57*, 417–422.

*Knott, V. E., Dollard, M. F., & Winefield, A. H. (2004). *Workplace bullying: An assessment of prevalence/nature of bullying behaviours in a correctional services sample*. Adelaide: Work & Stress Research Group, The University of South Australia.

Kokubun, S. (2007). Abusive behavior at work: A cross-cultural comparison between the United States and Japan (Doctoral dissertation, Alliant International University, San Diego, CA), Dissertation Abstracts International B 68/07, January 2008. ProQuest document ID: 1383483851.

*Kwok, R. P. W., Law, Y. K., Li, K. E., Ng, Y. C., Cheung, M. H., Fung, V. K. P., … Leung, W. C. (2006). Prevalence of workplace violence against nurses in Hong Kong. *Hong Kong Medical Journal*, *12*, 6–9.

Lapierre, L. M., Spector, P. E., & Leck, J. D. (2005). Sexual versus nonsexual workplace aggression and victims' overall job satisfaction: A meta-analysis. *Journal of Occupational Health Psychology*, *10*(2), 155–169.

*Lee, R. T., & Brotheridge, C. M. (2006). When prey turns predatory: Workplace bullying as a predictor of counteraggression/bullying, coping, and well-being. *European Journal of Work and Organizational Psychology*, *15*(33), 352–377.

Lewis, D. (1999). Workplace bullying – interim findings of a study in further and higher education in Wales. *International Journal of Manpower*, *20*(1/2), 106–118.

Lewis, D. (2004). Bullying at work: The impact of shame among university and college lectures. *British Journal of Guidance and Counselling*, *32*(3), 281–299.

*Lewis, D., & Gunn, R. (2007). Workplace bullying in the public sector: Understanding the racial dimension. *Public Administration*, *85*(3), 641–665.

Leymann, H. (1990). *Handbok för användning av LIPT-formuläret för kartläggning av risker för psykologisk vald [Manual of the LIPT questionnaire for assessing the risk of psychological violence at work]*. Stockholm: Violen.

*Leymann, H. (1992). *Vuxenmobbning på svenska arbetsplatser. En rikstäckande undersökning med 2438 intervjuer [Bullying among adults at Swedish workplaces. A nationwide survey with 2,438 inteviews]*. Stockholm: Arbetarskyddsstyrelsen.

Leymann, H. (1993). *Mobbning och utslagning i arbetslivet [Bullying and expulsion at work]*. Stockholm: Arbetsmiljöfonden.

Leymann, H. (1996). The content and development of mobbing at work. *European Journal of Work and Organizational Psychology*, *5*, 165–184.

*Leymann, H., & Tallgren, U. (1989). Undersöking av frekvensen vuxenmobbning inom SSAB med ett nytt formulär [A study on the prevalence of adult bullying in the SSAB with a new inventory]. *Arbete, människa, miljö*, *1*, 3–12.

Lipsey, M. W., & Wilson, D. B. (2001). *Practical meta-analysis* (Vol. 49). Thousand Oaks, CA: Sage.

*Lutgen-Sandvik, P., Tracy, S. J., & Alberts, J. K. (2007). Burned by bullying in the American workplace: Prevalence, perception, degree, and impact. *Journal of Management Studies*, *44*(6), 837–862.

*Magerøy, N., Lau, B., Riise, T., & Moen, B. E. (2009). Association of psychosocial factors and bullying at individual and departments levels among naval military personnel. *Journal of Psychosomatic Research*, *66*(4), 343–351.

Magley, V. J., Hulin, C. L., Fitzgerald, L. F., & DeNardo, M. (1999). Outcomes of self-labeling sexual harassment. *Journal of Applied Psychology*, *84*(3), 390–402.

*Malinauskiene, V., Obelenis, V., & Dopagienë, D. (2005). Psychological terror at work and cardiovascular diseases among teachers. *Acta Medica Lituanica*, *12*(2), 20–25.

*Martínez, M., Arroyo, J., Rodríguez, E., Seijo, R., Sepúlveda, M., & Tirado, M. (2002). *El acoso psicológico en el contexto laboral puertorriqueño: Un estudio exploratorio [Workplace mobbing in the Puerto Rican context: An exploratory study]*. Paper presented at the Hominis '02: Intercontinental Convention of Psychology and Human Sciences.

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

*Martínez, M., Guzmán, N., Martínez, M., Merle, J., Torres, W., & Viera, I. (2003). *El acoso psicológico en una muestra de empleados públicos en Puerto Rico [Workplace mobbing in a sample of public sector employees in Puerto Rico]*. Paper presented at the 50th Annual Convention of the Puerto Rico Psychology Association.

*Mathisen, G. E., Einarsen, S., & Mykletun, R. (2008). The occurrences and correlates of bullying and harassment in the restaurant sector. *Scandinavian Journal of Psychology*, *49*, 59–68.

Matthiesen, S. B., & Einarsen, S. (2001). MMPI-2 configurations among victims of bullying at work. *European Journal of Work and Organizational Psychology*, *32*, 335–356.

*Matthiesen, S. B., Raknes, B. I., & Røkkum, O. (1989). Mobbing på arbeidsplassen [Workplace bullying]. *Tidsskrift for Norsk Psykologforening*, *26*, 761–774.

*Mikkelsen, E. G., & Einarsen, S. (2001). Bullying in Danish worklife: Prevalence and health correlates. *European Journal of Work and Organizational Psychology*, *10*, 393–414.

Mullen, P. E., Pathé, M., Purcell, R., & Stuart, G. W. (1999). Study of stalkers. *American Journal of Psychiatry*, *156*, 1244–1249.

Neuman, J. H., & Baron, R. A. (2003). Social antecedents of bullying: A social interactionist perspective. In S. Einarsen, H. Hoel, D. Zapf, & C. L. Cooper (Eds.), *Bullying and emotional abuse in the workplace. International perspectives in research and practice* (pp. 185–202). London: Taylor & Francis.

*Neyens, I., Baillien, E., Vanoirbeek, K., & De Witte, H. (2005). *Risicofactoren voor het slachtofferschap van geweld, pesterijen en ongewenst seksueel gedrag in kleine en middelgrote organisaties. Een kwantitatieve studie [Risk factors for violence, bullying and sexual harassment at work within small- and medium sized organizations. A quantitative study]*. Leuven: Departement Psychologie.

*Niedhammer, I., David, S., & Degioanni, S. (2006). Economic activities and occupations at high risk for workplace bullying: Results from a large-scale cross-sectional survey in the general working population in France. *International Archives of Occupational and Environmental Health*, *80*(4), 346–353.

*Niedl, K. (1995). *Mobbing/Bullying am arbeitsplatz. Eine empiriche Analyse zum Phänomen sowie zu personalwirtschaftlich relevanten Effekten von systematischen Feindseligkeiten [Mobbing/bullying at work. An empirical analysis of the phenomenon and of the effects of systematic harassment on human resource management]*. Munich: Hampp.

*Niedl, K. (1996). Mobbing and well-being: Economic and personnel development implications. *European Journal of Work and Organizational Psychology*, *5*(2), 239–249.

Nielsen, M. B., & Einarsen, S. (2008). Sampling in research on interpersonal aggression. *Aggressive Behavior*, *34*(3), 265–272.

*Nielsen, M. B., Skogstad, A., Matthiesen, S. B., Glasø, L., Aasland, M. S., Notelaers, G., & Einarsen, S. (2009). Prevalence of workplace bullying in Norway: Comparisons across time and estimation methods. *European Journal of Work and Organizational Psychology*, *18*(1), 81–101.

*Normann, T. M., & Rønning, E. (2007). *Konflikter, mobbing, trakassering og vold i arbeidslivet. Få utsatt, noen yrker mer utsatt enn andre [Conflicts, bullying, harassment and violence in the working life. Few exposed, some occupations more exposed than other]*. Oslo: Statistisk Sentralbyrå/Statistics Norway.

*Notelaers, G., & De Witte, H. (2003). *De relatie tussen werkstress en pesten op het werk en welbevinden op het werk [The relationship between work stressors, workplace bullying and well-being]*. Reeks: De Arbeidsmarkt in Vlaanderen. Leuven, Steunpunt Werkgelegenheid, Arbeid en Vorming 26 mei 2003, VUB.

*Notelaers, G., De Witte, H., Vermunt, J., & Einarsen, S. (2006). Pesten op het werk, gewikt en gewogen: Een latente klassen benadering op basis van de negatieve acts-vragenlijst [How to measure bullying at work? A latent class analysis of the Negative Acts Questionnaire]. *Gedrag en Organisatie*, *19*(2), 149–160.

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

Notelaers, G., Einarsen, S., De Witte, H., & Vermunt, J. (2006). Measuring exposure to bullying at work: The validity and advantages of the latent class cluster approach. *Work and Stress*, *20*(4), 288–301.

Nunnally, J., & Bernstein, I. H. (1994). *Psychometric theory* (3rd ed.). New York: Wiley.

*O'Connell, P. J., Calvert, E., & Watson, D. (2007). *Bullying in the workplace: Survey reports, 2007*. Dublin: The Economic and Social Research Institute.

*O'Connell, P. J., & Williams, J. (2002). *The incidence and correlates of workplace bullying in Ireland*. Dublin: Economic and Social Research Institute.

O'Donohue, W., Downs, K., & Yeater, E. A. (1998). Sexual harassment: A review of the literature. *Aggression and Violent Behavior*, *3*(2), 111–128.

*Ólafsson, R. F., & Jóhannsdóttir, H. L. (2004). Coping with bullying in the workplace: The effect of gender, age and type of bullying. *British Journal of Guidance and Counselling*, *32*(3), 319–333.

Olweus, D. (1991). Bullying/victim problem among school children. In I. Rubin & D. Pepler (Eds.), *The development and treatment of childhood aggression* (pp. 411–448). Hillsdale, NJ: Erlbaum.

O'Moore, A. M., & Hillery, B. (1989). Bullying in Dublin schools. *Irish Journal of Psychology*, *10*, 426–441.

*O'Moore, M., Lynch, J., & Niamh, N. D. (2003). The rates and relative risks of workplace bullying in Ireland, a country of high economic growth. *International Journal of Management and Decision Making*, *4*(1), 82–95.

*Ortega, A., Høgh, A., Pejtersen, J. H., & Olsen, O. (2008). Prevalence of workplace bullying and risk groups: A representative population study. *International Archives of Occupational and Environmental Health*, *82*(3), 417–426.

Østvik, K., & Rudmin, F. (2001). Bullying and hazing among Norwegian army soldiers. Two studies of prevalence, context, and cognition. *Military Psychology*, *13*, 17–39.

*Out, J. (2005). *Meanings of workplace bullying: Labelling versus experiencing and the belief in a just world*. Unpublished Dissertation, University of Windsor, Windsor, Ontario, Canada.

*Paice, E., Aitken, M., Houghton, A., & Firth-Cozens, J. (2004). Bullying among doctors in training: Cross sectional questionnaire survey. *British Medical Journal*, *329*, 658–659.

Palmer, E. J., & Thakordas, V. (2005). Realtionship between bullying and scores on the Buss–Perry Aggression Questionnaire among imprisoned male offenders. *Aggressive Behavior*, *31*, 56–66.

Parkins, I. S., Fishbein, H., & Ritchey, P. N. (2006). The influence of personality on workplace bullying and discrimination. *Journal of Applied Social Psychology*, *36*(10), 2554–2577.

*Perkiö-Mäkelä, M., Hirvonen, M., Elo, A., Ervasti, J., Huuhtanen, P., Kandolin, I., … Reijula, K. (2006). *Työ ja terveys – haastattelututkimus 2006. Taulukkoraportti [Work and health interview survey 2006. Table report]*. Helsinki: Finnish Institute of Occupational Health.

*Piirainen, H., Elo, A.-L., Hirvonen, M., Kauppinen, K., Ketola, R., Laitinen, H., … Virtanen, S. (2000). *Työ ja terveys – haastattelututkimus 2000. Taulukkoraportti [Work and health interview survey 2000. Table report]*. Helsinki: Finnish Institute of Occupational Health.

*Piirainen, H., Elo, A.-L., Kankaanpää, E., Laitinen, H., Lindström, K., Luopajärvi, T., … Riala, R. (1997). *Työ ja terveys – haastattelututkimus 1997. Taulukkoraportti [Work and health interview survey 1997. Table report]*. Helsinki: Finnish Institute of Occupational Health.

*Piirainen, H., Hirvonen, M., Elo, A.-L., Huuhtanen, P., Kandolin, I., Kauppinen, K., … Virtanen, S. (2003). *Työ ja terveys – haastattelututkimus 2003. Taulukkoraportti [Work and health interview survey 2003. Table report]*. Helsinki: Finnish Institute of Occupational Health.

*Piñuel, I. (2006). *Mobbing, Acoso psicológico en el trabajo [Mobbing, psychological harassment at work]*. Madrid: Instituto Regional de Seguridad Y Salud en eL Trabajo Conserjería de Empleo Y Mujer.

Power, K. G., Dyson, G. P., & Wozniak, E. (1997). Bullying among Scottish young offenders: Inmates' self-reported attitudes and behaviour. *Journal of Community and Applied Social Psychology*, *7*, 209–218.

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

*Privitera, C., & Campbell, M. A. (2009). Cyberbullying: The new face of workplace bullying. *CyberPsychology and Behavior*, *12*(4), 395–400.

Quine, L. (1999). Workplace bullying in NHS community trust: Staff questionnaire survey. *British Medical Journal*, *318*, 228–232.

*Quine, L. (2002). Workplace bullying in junior doctors: Questionnaire survey. *British Medical Journal*, *324*, 878–879.

*Raknes, B. I., Einarsen, S., Matthiesen, S. B., & Hellesøy, O. H. (1991). Mobbing på arbeidsplassen – et helseproblem? *Norsk Tidsskrift for Arbeidsmedisin*, *12*(3), 155–160.

*Rayner, C. (1997). The incidence of workplace bullying. *Journal of Community and Applied Social Psychology*, *7*, 199–208.

*Rayner, C. (1999). From research to implementation: Finding leverage for prevention. *International Journal of Manpower*, *20*(1/2), 28–38.

Rayner, C., Hoel, H., & Cooper, C. L. (2002). *Workplace bullying. What we know, who is to blame, and what can we do?* London: Taylor & Francis.

Richman, J. A., Flaherty, J. A., & Johnson, T. P. (1999). Sexual harassment and generalized workplace abuse among university employees: Prevalence and mental health correlates. *American Journal Of Public Health*, *89*(3), 358–363.

*Rodríguez, M., & Martínez, M. (2003). *El acoso psicológico en el trabajo en una muestra de empleados de organizaciones de servicios de salud [Workplace mobbing in a sample of health services organizations]*. Paper presented at the 50th Annual Convention of the Puerto Rico Psychology Association.

*Salin, D. (2001). Prevalence and forms of bullying among business professionals. A comparison of two different strategies for measuring bullying. *European Journal of Work and Organizational Psychology*, *10*(4), 425–441.

Saunders, P., Huynh, A., & Goodman-Delahunty, J. (2007). Defining workplace bullying behaviour professional lay definitions of workplace bullying. *International Journal of Law and Psychiatry*, *30*(4–5), 340–354.

Schat, A. C., Frone, M. R., & Kelloway, E. K. (2006). Prevalence of workplace aggression in the U.S. workforce: Findings from a national study. In E. K. Kelloway, J. Barling, & J. J. Hurrell (Eds.), *Handbook of workplace violence* (pp. 47–89). Thousand Oaks, CA: Sage.

Schat, A. C., & Kelloway, E. K. (2004). Workplace aggression. In J. Barling, M. R. Frone, & E. K. Kelloway (Eds.), *Handbook of work stress* (pp. 198–218). Thousand Oaks, CA: Sage.

Schulze, R. (2004). *Meta-analysis. A comparison of approaches*. Göttingen: Hogrefe & Hubert Publishers.

Smith, B. H., Hannaford, P., Elliot, A. M., Smith, W. C., & Chambers, W. A. (2005). The 'number needed to sample' in primary care research. Comparison of two primary care sampling frames for chronic back pain. *Family Practice*, *22*, 205–214.

*Steinman, S. (2003). *Workplace violence in the health sector. Country case study: South Africa*. Geneva: ILO/ICN/WHO/PSI Joint Programme on Workplace Violence in the Health Sector.

*Tehrani, N. (2004). Bullying: A source of chronic post traumatic stress. *British Journal of Guidance and Counselling*, *32*, 358–366.

*Thomas, M. (2004). Bullying among support staff in a higher education institution. *Health Education*, *105*(4), 273–288.

*Tuckey, M. R., Dollard, M. F., Hosking, P. J., & Winefield, A. H. (2009). Workplace bullying: The role of psychosocial work environment factors. *International Journal of Stress Management*, *16*(3), 215–232.

van Beest, I., & Williams, K. D. (2006). When inclusion costs and ostracism pays, ostracism still hurts. *Journal of Personality and Social Psychology*, *91*(5), 918–928.

*Varhama, L. M., & Björkqvist, K. (2004a). Conflicts, burnout, and bullying in a Finnish and a Polish company: A cross national comparision. *Perceptual and Motor Skills*, *98*, 1234–1240.

*Varhama, L. M., & Björkqvist, K. (2004b). Conflicts, workplace bullying and burnout among problems among municipal employees. *Psychological Reports*, *94*, 1116–1124.

Copyright © The British Psychological Society
Reproduction in any form (including the internet) is prohibited without prior permission from the Society

*Vartia, M. (1996). The sources of bullying – psychological work environment and organizational climate. *European Journal of Work and Organizational Psychology*, *5*, 203–214.

*Vartia, M., & Hyyti, J. (2002). Gender differences in workplace bullying among prison officers. *European Journal of Work and Organizational Psychology*, *11*(1), 113–126.

Viswesvaran, C., & Schmidt, F. L. (1992). A meta-analytic comparison of the effectiveness of smoking cessation methods. *Journal of Applied Pscyhology*, 77(4), 554–561.

Voss, M., Floderus, B., & Diderichsen, F. (2001). Physical, psychological, and organisational factors relative to sickness absence: A study based on Sweden post. *Occupational and Environmental Medicine*, *58*, 178–184.

Walsh, B. R., & Clarke, E. (2003). Post-trauma symptoms in health workers following physical and verbal aggression. *Work and Stress*, *17*(2), 170–181.

Willness, C. R., Steel, P., & Lee, K. (2007). A meta-analysis of the antecedents and consequences of workplace sexual harassment. *Personnel Psychology*, *60*, 127–162.

Zapf, D., Einarsen, S., Hoel, H., & Vartia, M. (2003). Empirical findings on bullying in the workplace. In S. Einarsen, H. Hoel, D. Zapf, & C. L. Cooper (Eds.), *Bullying and emotional abuse in the workplace. International perspectives in research and practice* (pp. 103–126). London: Taylor & Francis.

Received 27 March 2009; revised version received 10 November 2009